

Discriminator Validity & Generalization — v4

2026-07-27 · Companion to RESULTS_BD_V4.pdf . Question under test: **does the unseen (e99) discriminator genuinely generalize to a new scenario, or does the downstream seen=unseen equality hide something?** All discs here are the v4 round (_v4), trained on the EOS-fixed backbones' pools; the pre-fix (v2) numbers appear only in the confound case study (§4).

1. What the discriminator is, and split hygiene

- Role:** guidance needs the block-conditional ratio $w = q_{\text{except}}(x_0|x^{<b})/p_\theta(x_0|x^{<b})$; the disc D estimates it as $D/(1-D)$ on clean partial paths [prefix | block], adjacency-conditioned (GraphSAGE embedding of A_{scn} + per-position edge-existence features), bounded logit $\ell = 4 \cdot \tanh(z/4)$.
- Seen (e0):** positives = except_0 paths, 1% of the pool *after excluding* the 1,000 eval-reserved rows (RandomState(777) permutation, first 1,000 reserved). **Unseen (e99):** positives = except 1–99 only (per-scenario 1%), except_0 never contributes a gradient; the except_0 validation accuracy printed during training is a monitoring diagnostic only (fixed 4,000 steps, no model selection).
- Every evaluation below uses held-out data for both discs:** except_0 probes use the reserved rows; except_1 probes use rows outside e99's 1% slice; negatives are fresh (seed-99 pools disjoint from the seed-1 training pools; fresh normal sample for the data-negative probe).

2. Symmetric probe design

Two probe scenarios flip the roles: on **except_0** the e0 disc is in-scenario and e99 is zero-shot; on **except_1** e99 is in-scenario (held-out rows) and e0 is zero-shot. If generalization is real, the in-scenario advantage should be small and should flip with the probe — which is exactly what happens. Metrics: AUC, accuracy@0, and Pearson correlation between the two discs' logits on identical inputs.

3. Model-negative discs (per block): positives = real except paths, negatives = fresh unconditional generations

3.1 fam 0.05

blk	AUC on except_0 (e0 in-scn)		AUC on except_1 (e99 in-scn)		logit Pearson	
	e0	e99	e0	e99	exc0	exc1
1	0.870	0.833	0.787	0.848	0.904	0.810
2	0.890	0.872	0.843	0.887	0.913	0.837
4	0.897	0.888	0.850	0.901	0.925	0.825
8	0.943	0.931	0.898	0.932	0.956	0.866
16	0.951	0.948	0.930	0.948	0.970	0.947
32	0.956	0.951	0.945	0.952	0.963	0.950
64	0.957	0.958	0.946	0.957	0.976	0.968

3.2 fam 0.1

blk	AUC on except_0 (e0 in-scn)		AUC on except_1 (e99 in-scn)		logit Pearson	
	e0	e99	e0	e99	exc0	exc1
1	0.947	0.935	0.902	0.926	0.950	0.877
2	0.959	0.953	0.914	0.940	0.963	0.920
4	0.962	0.948	0.929	0.946	0.970	0.933
8	0.977	0.971	0.946	0.965	0.975	0.929
16	0.981	0.978	0.962	0.972	0.980	0.959
32	0.977	0.973	0.964	0.971	0.981	0.965
64	0.983	0.977	0.967	0.974	0.980	0.973

- The in-scenario advantage flips with the probe direction and stays small** (0.005–0.06 AUC): whichever disc is zero-shot loses a little, in both directions, at both families. This is the signature of genuine transfer, not memorization.
- Transfer is cheap here by construction:** on this task most of the signal — edge legality against the *input* adjacency, plus model-generation artifacts — is scenario-agnostic. The disc reads A_{scn} as an input, so "an edge absent from the given adjacency $\Rightarrow$ fake" transfers to any scenario.

- **AUC level across blocks tracks the backbone, not the disc:** it rises with the negative pool's invalid-edge rate (v4: blk1 2.3% → blk64 11.4%). Cross-block AUC comparison is a statement about p_0 quality (harder vs easier negatives), not about disc quality — see §4.
- fam 0.1 AUCs are uniformly higher: twice the removed edges give a stronger scenario signal.

4. The negative-pool confound — v2 blk1 case study

Under the pre-fix (v2) backbones, the blk1 disc posted the *best* AUC (0.985) while blk2 posted the worst (0.881) — an apparent non-monotonicity. The cause was the negative pool, not the disc: v2 blk1's unconditional generations were degenerate (every path ran to the canvas cap, length 101 vs real 23.2; 50.5% invalid edges), so its negatives were separable by a **length shortcut** alone. After the EOS fix the blk1 pool is healthy (length 25.2; 2.3% invalid) and the blk1 AUC falls to an honest 0.870 — now the AUC sequence is monotone in the pools' invalid-edge rates at every block. Lesson: with model-negative training, **per-block AUCs conflate disc quality with generation quality**; only the within-block seen-vs-unseen comparison (same pool for both discs) is confound-free.

	pool len (real 23.2)	pool invE	cap%	AUC e0, exc0
v2 blk1 (pre-fix)	101.0±0.0	50.5%	100%	0.985 (<i>length shortcut</i>)
v4 blk1 (post-fix)	25.2±12.5	2.3%	0.9%	0.870 (<i>honest</i>)

5. Data-negative discs (block-independent): the strict generalization probe

Positives = real except paths, negatives = **real normal paths** — real-vs-real, so every scenario-agnostic shortcut (model artifacts, length) is removed and **only scenario-specific information** (which edges were removed, how routes divert) separates the classes. One disc per regime, no per-block pools, hence no §4 confound. These discs involve no backbone at all, so they are identical across v2/v4 (same seed).

family	AUC on except_0 (e0 in-scen)		AUC on except_1 (e99 in-scen)		zero-shot drop		Pearson exc0 / exc1
	e0	e99	e0	e99	e0	e99	
0.05	0.845	0.813	0.745	0.835	-0.100	-0.022	0.913 / 0.737
0.1	0.951	0.932	0.867	0.914	-0.084	+0.018	0.956 / 0.834

- **This probe separates the training regimes:** the single-scenario e0 disc degrades substantially zero-shot (-0.100 / -0.084), while the 99-scenario e99 disc is nearly flat (-0.022 / +0.018). Genuine scenario-specific generalization is a property of *multi-scenario training* — one scenario does not teach the adjacency-conditioning rule well enough.
- The logit correlation also drops on the zero-shot side (exc1: 0.737 / 0.834) — where the task is scenario-specific, the two discs really are different functions.

6. Why downstream seen ≈ unseen even so

- **Guidance uses the model-negative disc**, whose signal transfers almost freely (§3) — the hard probe of §5 concerns a disc variant the pipeline does not deploy (it is the §4.1b ablation in the v3 report).
- **Self-normalized IS cancels calibration:** weights $w_n = \exp(\psi \ell_n)$ enter only as $w_n / \sum w_m$, so any constant logit offset — the main casualty of zero-shot transfer (accuracy@0 drops much more than AUC) — multiplies out exactly. Only the ranking/relative differences matter, and those transfer (AUC gap ≤0.05).
- **At guidance time the scored inputs are model candidates**, i.e. exactly the distribution the model-negative disc was trained against; the deployed disc is matched to its use.

7. Conclusion

The seen/unseen equivalence observed downstream (v↔a gap ≤0.015 at fam 0.05, ≤0.003 at fam 0.1 across all blocks) is fully accounted for: (1) the deployed model-negative discriminators transfer because their discrimination signal is scenario-agnostic by construction and adjacency is an input; (2) SNIS guidance is invariant to the calibration shift that zero-shot transfer does cause; and (3) independently, the strict real-vs-real probe shows the e99 training recipe does acquire transferable scenario-specific knowledge that single-scenario training does not. The one caveat for reading disc metrics — per-block AUCs reflect negative-pool difficulty — is demonstrated and resolved by the v2→v4 blk1 case study.

Sources: `disc_gen_eval.py` ; JSONs `sets_res/disc_gen_eval_f{005,01}{,_data}_v4.json` ; pre-fix contrast `disc_gen_eval_f005.json` (v2 round); pool stats `uncond_pool_eval{,_v4}.json` .